

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Indicative content: RELIABLE The power stations are all joined by a network of cables so that if any one of them fails or is off-line for maintenance, or there is a sudden surge in demand, then others can make up for the deficit e.g. pumped storage systems can be used. Electrical power can also be imported from abroad for this purpose. EFFICIENT The efficiency of the grid is high because the electricity that is generated is increased to a very high voltage by a step-up transformer at the power station for transmission along the grid cables. This reduces current in the cables and minimises energy losses due to heating, improving efficiency. SAFE The high voltage is finally reduced in a step-down transformer before being supplied to factories, towns, schools and homes to make it safer to use. Pylons support cables high above the ground. 5-6 marks Comprehensive description of all 3 areas i.e. reliability, efficiency and safety. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Comprehensive description of 2 out of 3 areas i.e. reliability, efficiency and safety or limited description of all 3 areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Comprehensive description of 1 out of 3 areas i.e. reliability, efficiency and safety or limited description of 1 or 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> <i>The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						
	(b)	(i)		Sweden		1		1		
		(ii)		Norway uses more heating / lighting (1) because they get longer winters or colder or darker (1) Don't accept more people			2	2		
		(iii)		Units used in Norway = 4 times as much as in U.K. (1) Cost in Norway = 2 times its cost in the U.K. (1) so disagree To award both marks the conclusion must be present			2	2	2	
		(iv)	I.	$3 \times 5 \times 7$ (1) = 105 [kWh] (1) Answer of 15 OR 21 OR 35 seen anywhere award 1 mark	1	1		2	2	
			II.	Cost = 105 ecf $\times 8$ (1) = 840 [p] (1) Award 1 mark only for 105 ecf $\times 0.08 = 8.40$	1	1		2	2	
				Question 3 total	8	3	4	15	6	0